Individual Task 1: Part 1

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This paper explores the potential of using supervised machine learning to aid in decision-making for credit risks in retail finance. It describes and analyses two publicly available consumer credit data sets, which are examined for information leakage post decision and models using Decision Tree and Support Vector Machine. Results are evaluated using ROC-AUC, precision, recall, and F1 measures.

Overleaf project URL: https://www.overleaf.com/read/REPLACE-WITH-YOUR-PROJECT

1 Data Science Role

1.1 Job Description and Role Identification

The job I have selected for this assignment is a **Data Scientist** role advertised on SEEK by **Bluefin Resources Pty Limited**, recruiting on behalf of a fintech client in consumer lending. It is a full-time hybrid position based in Melbourne, listed under Banking & Financial Services – Analysis & Reporting, with a salary range of AUD $130,000–$150,000.

The role focuses primarily on model development and deployment. The position involves building machine learning models for credit risk and fraud, developing predictive scoring models using large transactional and customer datasets, and working with risk, engineering, and product teams to translate the model output into practical business decision rules. The advertised skills include Python, SQL, supervised machine learning, and experience with tree-based models.

The reason behind choosing this specific job role is because it aligns well with the data science work presented in this report. The models developed in Section 3 focus on predicting loan default from application data, which directly relates to the credit-risk modelling described in the job advertisement.

Job listing URL: https://au.seek.com/job/93510276?ref=search-standalone&type=standard&origin=jobTitle#sol=bb556804a559a5c8c51c1c3859ddcbed29e00238

1.2 Industry Context

Retail banking and consumer lending use data science to assess the likelihood of missed repayments. It supports better lending decisions, fraud detection, loan pricing, risk management, and compliance with responsible lending and capital requirements.

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1.3 Role Focus

Prediction-focused mainly. This job involves working in Home Buying Credit Risk area where one develops machine learning models to improve credit risk models and decisions. Insights are very important but secondary: explainability is essential since the model results go into automated approvals. The role is mainly prediction-focused, as it involves developing machine learning models to improve credit risk assessment and lending decisions. While generating useful insights is also important, explainability is essential because model results can directly influence automated credit approvals and other important lending decisions.

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1.4 Value Creation

Better ranking of applicants by repayment risk lets the bank approve more creditworthy borrowers it would otherwise decline, and decline or price appropriately those likely to struggle. That improves approval rates and margin, reduces impairment losses, and protects customers from loans they cannot sustain.
[Word count: 44]

1.5 Values and Culture Alignment

Learning, working together, and being accountable with data are important to me. This job suits my interests as it allows me to work with both technology and decision-making in fin-tech. It is also something I would like to do in an environment where collaboration and communication prevail. **[Word count: 49]**

1.6 Diversity and Inclusion

Credit models can learn from past decisions that may have disadvantaged certain groups. Diverse teams can question whether selected features indirectly represent factors such as location, gender, or migrant status, and identify groups that are underrepresented in the training data, helping produce fairer and more representative models.
[Word count: 47]

1.7 Cover Letter

Hiring Manager,

I would like to apply for the Data Scientist role that is listed by Bluefin Resources Pty Limited on behalf of its fintech client that operates in the domain of consumer lending. Currently, I am pursuing the Master's program in Data Science at RMIT University where I am acquiring skills in data analytics, machine learning, decision-making and, gradually gaining interest in finance sector specifically credit risk.

Previously, I was working with NIQ in India – global consumer intelligence company. The above experience enabled me to gain first-hand knowledge in the field of data, data quality, data analysis and reports, and transformation of information into business insights. Also, working with actual business data helped me to adopt a more practical approach towards resolving issues related to data.

In addition to the above-mentioned professional experience, I acquired skills in Python, SQL, data preprocessing, exploratory data analysis, and machine learning. As for now, my data science project is dedicated to the prediction of loan repayment risk based on two publicly available credit data sets. The aim of the project is to apply machine learning algorithms in order to discover the patterns correlated with default loans and analyze efficiency of such models in making credit-risk decisions.

I am very interested in this opportunity since it allows to combine technical modelling with the cooperation with risk, engineering and product teams. It is important for me to continue my learning, collaborate with others, communicate effectively and use data responsibly.

Currently, I am based in Melbourne and eager to continue my data science career in technology-based financial environment.

Thank you for your consideration.

Yours sincerely, Dhrumil Ashish Parikh

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2 Data Set

2.1 Description

Home Credit Default Risk [1] comprises loan applications from customers who have a poor credit history. Its major application table consists of **307,511** observations and **122** features that include income, employment status, housing, demographic and credit scores. Lending Club Credit Risk [?] includes **887,379** loans issued in the US using peer-to-peer lending platforms, with a total of **74** variables that consist of loan amounts, interest rates, loan grades, income levels, employment periods, debt-income ratios, and credit histories.

[Word count: 83]

Dataset 1: https://www.kaggle.com/competitions/home-credit-default-risk/overview

Dataset 2: https://www.kaggle.com/datasets/gabrieloliveirasan/lending-club-credit-risk

2.2 Why I have these datasets

The two datasets were chosen because they closely match the selected Bluefin role, which involves developing machine learning models for credit risk and consumer lending. Home Credit provides information on loan applications and repayment difficulties, while Lending Club provides a different lending environment for comparison. Using both separately allows me to investigate whether similar risk patterns can be identified across different credit datasets and assess how well the models support lending decisions.

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3 Part 1.3: Data Analysis

3.1 Preprocessing and Feature Engineering

Both data sets went through a standardized process of pre-processing. Features with more than 45% of missing values were dropped from the Home Credit Default Risk data set; numerical features were imputed with the median value and categorical features with the mode. A known data issue for `DAYS_EMPLOYED` (an outlier value of 365,243 which represented unemployment or retirement for the applicant) was marked with an indicator feature and not considered to be a real outlier, since it is common practice for this particular data set. For Lending Club, data records with an ambiguous loan status (i.e., `Current`, `In Grace Period`, `Late`, `Issued`) were dropped, because there was no known label for those loans yet.

Domain-informed features were engineered for both datasets, including credit-to-income ratio, annuity/installment-to-income ratio, and loan-to-income ratio, reflecting standard underwriting heuristics used in credit risk scoring. Categorical variables were one-hot encoded, and numeric features were standardized. No duplicate records were found in Home Credit; Lending Club also returned zero duplicates after restriction to resolved loans.

Both datasets exhibit substantial class imbalance, with defaults comprising only 8.1% of Home Credit applications and 18.7% of resolved Lending Club loans (Figure 1 shows both). This imbalance was addressed directly within each model rather than through resampling: `class_weight="balanced"` for Random Forest, and `scale_pos_weight` for XGBoost, both of which up-weight the minority (default) class during training without altering the underlying data distribution.

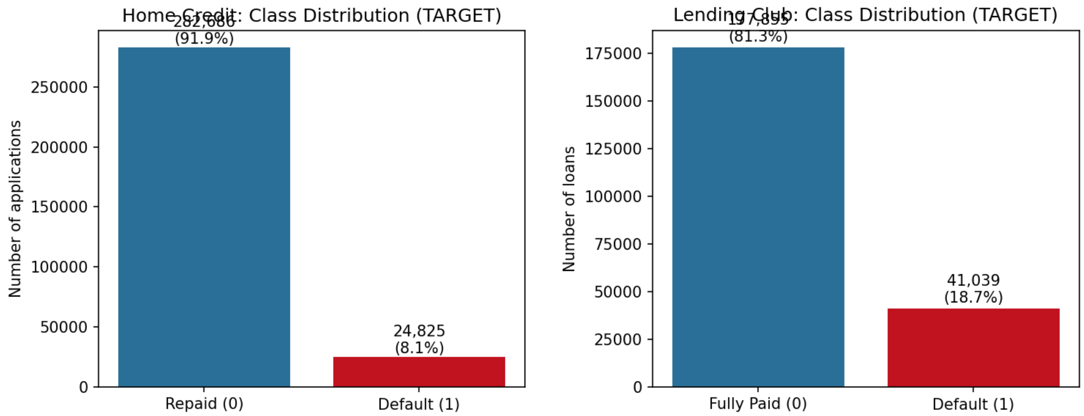

Fig. 1. Class distribution of the target variable for Home Credit (left) and Lending Club (right), confirming substantial imbalance toward the non-default class in both datasets.

Table 1. Dataset Summary

Attribute	**Home Credit Default Risk**	**Lending Club**
Original records	307,511	887,379
Records used for modelling	307,511	218,894 (resolved loans only)
Original attributes	122	74 (24 retained)
Post-encoding feature count	177	130
Default (minority class) rate	8.1%	18.7%
Duplicate records removed	0	0

Table 2. Selected Feature Descriptions

Feature	**Dataset**	**Description**
EXT_SOURCE_1/2/3	Home Credit	Normalised external credit bureau scores
CREDIT_INCOME_RATIO	Home Credit (engineered)	Loan amount relative to applicant income
DAYS_EMPLOYED_ANOM	Home Credit (engineered)	Flag for placeholder unemployment value
int_rate	Lending Club	Interest rate assigned to the loan
dti	Lending Club	Debt-to-income ratio
revol_util	Lending Club	Revolving credit utilisation percentage
loan_to_income	Lending Club (engineered)	Loan amount relative to annual income

3.2 Correlation Structure

Figures 2 and 3 show correlation heatmaps for a subset of key numeric features in each dataset. In Home Credit, the external bureau scores show a moderate negative correlation with default, consistent with their design as risk indicators. In Lending Club, interest rate shows the strongest positive correlation with default, which is expected since Lending Club's own grading system already prices perceived risk into the interest rate at origination.

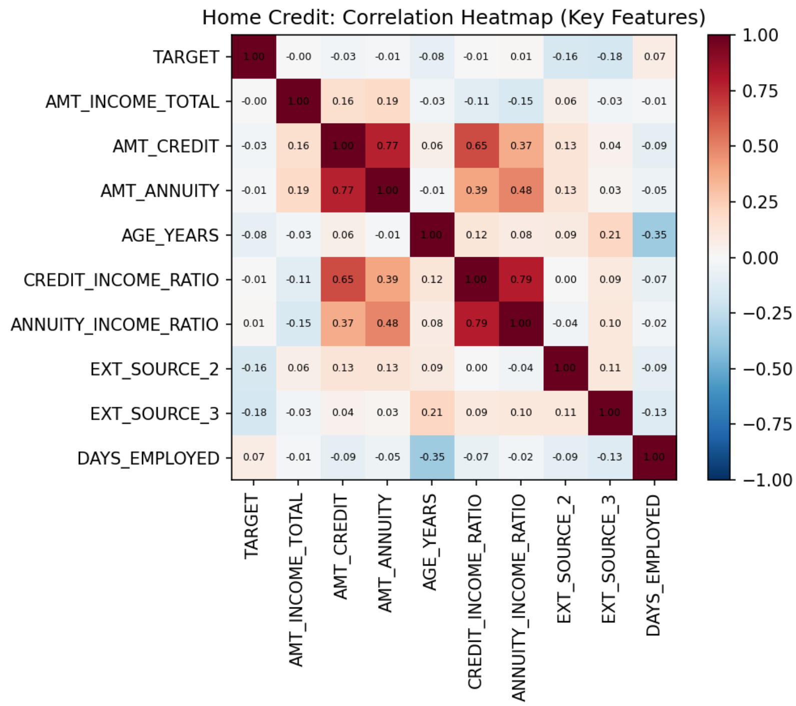

Fig. 2. Correlation heatmap of key Home Credit numeric features against the default target. External bureau scores (EXT_SOURCE_2/3) show the strongest negative association with default.

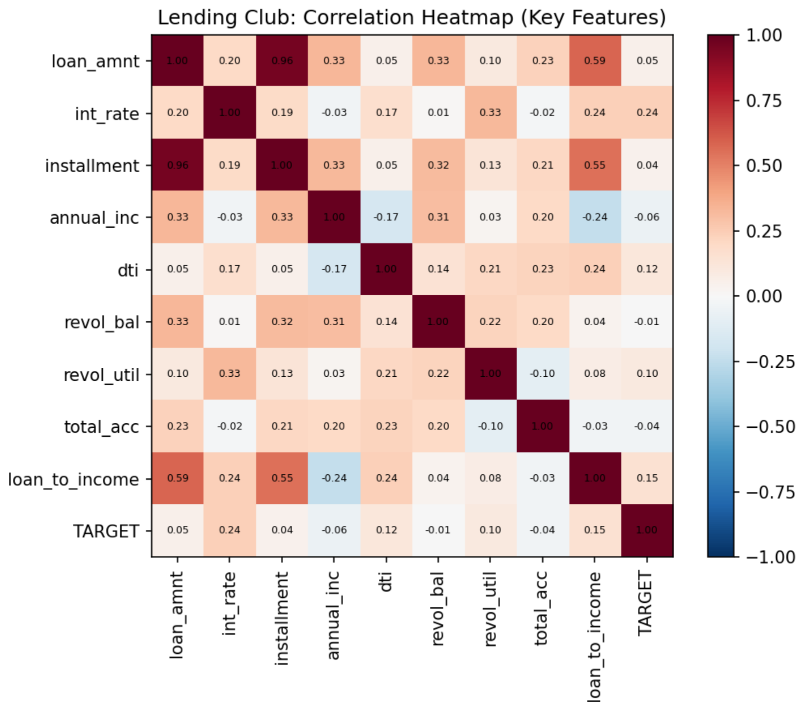

Fig. 3. Correlation heatmap of key Lending Club numeric features against the default target. Interest rate shows the strongest positive association with default.

3.3 Model Training

Two supervised machine learning models were implemented for each dataset: **Random Forest** and **XGBoost**. Both algorithms use decision trees; however, they differ in implementation and operation. Random Forest creates many trees and then aggregates their decisions, whereas XGBoost sequentially creates new trees aimed at improving the prediction of errors made by the previous tree.

The datasets were divided into 80% training and 20% testing subsets with stratification to preserve the class distribution. Random Forest had 300 trees with depths ranging from 12 to 14 and minimum leaf sizes from 20 to 30, which helped prevent overfitting in the case of one-hot encoding. XGBoost used 400 estimators, a learning rate of 0.05, maximum tree depth of 5, and sampling parameters of 0.8 to decrease model complexity.

3.4 Results and Evaluation

Because of the strong imbalance in both data sets, accuracy cannot be used as an evaluation measure [**?**]: a classifier that classifies all samples to the majority class would have an accuracy of 91.9% on the Home Credit data set while recognizing no defaulter, which is not useful at all from an operational point of view for the fraud/risk department. The measures of interest will be thus ROC-AUC (the ranking capability across all decision thresholds), precision, recall, and F1-score (the performance at the decision threshold), and average precision.

Table 3. Model Comparison – Home Credit Default Risk

Model	ROC-AUC	Precision	Recall	F1-score	Avg. Precision
Random Forest	0.743	0.179	0.579	0.274	0.219
XGBoost	**0.765**	0.175	**0.668**	**0.277**	**0.252**

Table 4. Model Comparison – Lending Club

Model	ROC-AUC	Precision	Recall	F1-score	Avg. Precision
Random Forest	0.709	0.308	0.624	0.412	0.360
XGBoost	**0.720**	**0.311**	**0.656**	**0.422**	**0.375**

XGBoost outperformed Random Forest on every metric across both datasets. This is consistent with the boosting mechanism: because each successive tree targets the residual errors left by the previous ensemble, XGBoost can allocate additional model capacity specifically to the hard-to-classify borderline cases that dominate error in imbalanced classification, whereas Random Forest's independently grown trees do not share this corrective feedback loop. The gap is more pronounced on Home Credit (ROC-AUC +0.022, average precision +0.033) than on Lending Club (ROC-AUC +0.011), suggesting XGBoost's advantage is larger when the feature set is noisier and more heterogeneous, as is the case with Home Credit's 177 encoded features spanning demographic, housing, and document-flag variables.

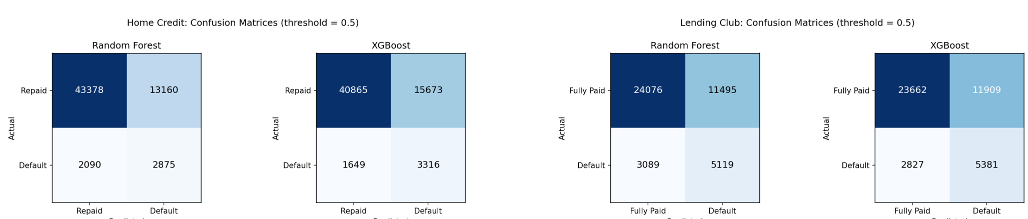

Fig. 4. Confusion matrices at the default 0.5 probability threshold for Home Credit (left) and Lending Club (right). Both models trade off precision for higher recall on the default class as a deliberate consequence of class-weighting, catching more true defaulters at the cost of more false positives.

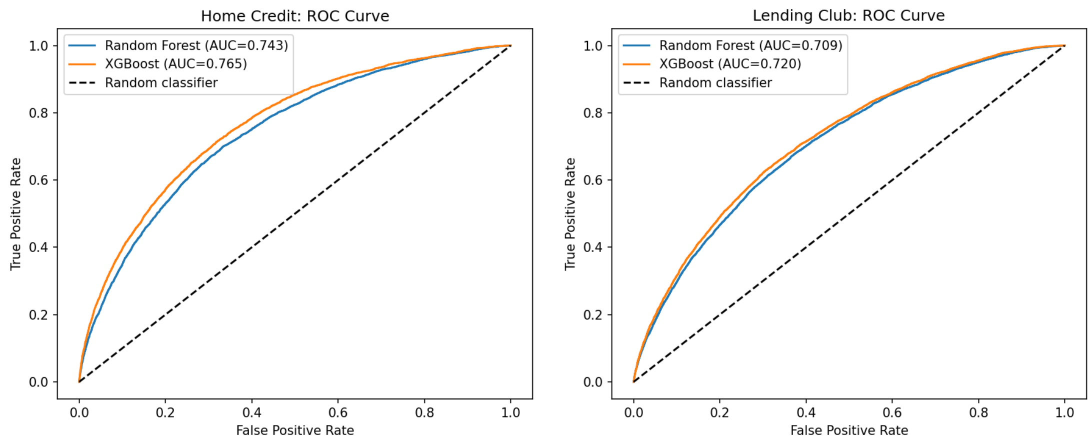

Fig. 5. ROC curves for Home Credit (left) and Lending Club (right). XGBoost dominates Random Forest across nearly the full range of thresholds on both datasets.

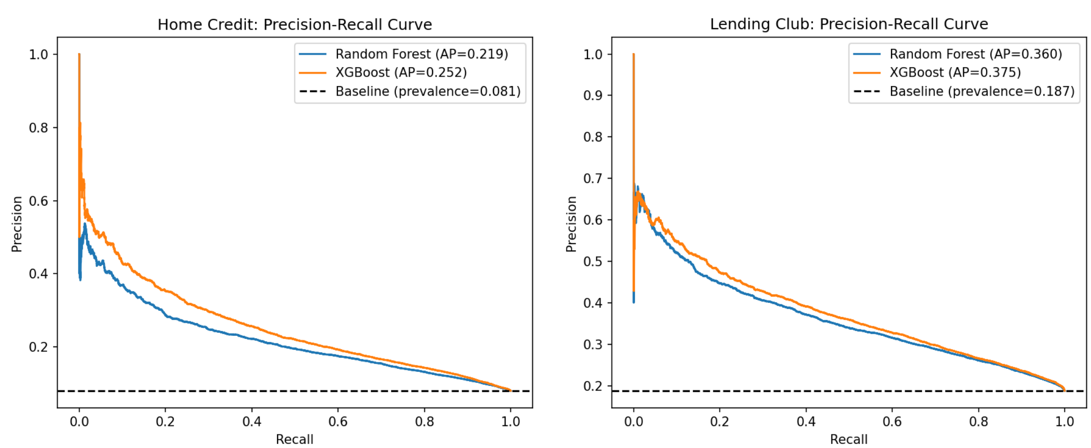

Fig. 6. Precision-recall curves for Home Credit (left) and Lending Club (right), shown against the no-skill baseline equal to class prevalence. Both models substantially outperform random guessing, though absolute precision remains modest, reflecting the inherent difficulty of the classification problem at low base rates.

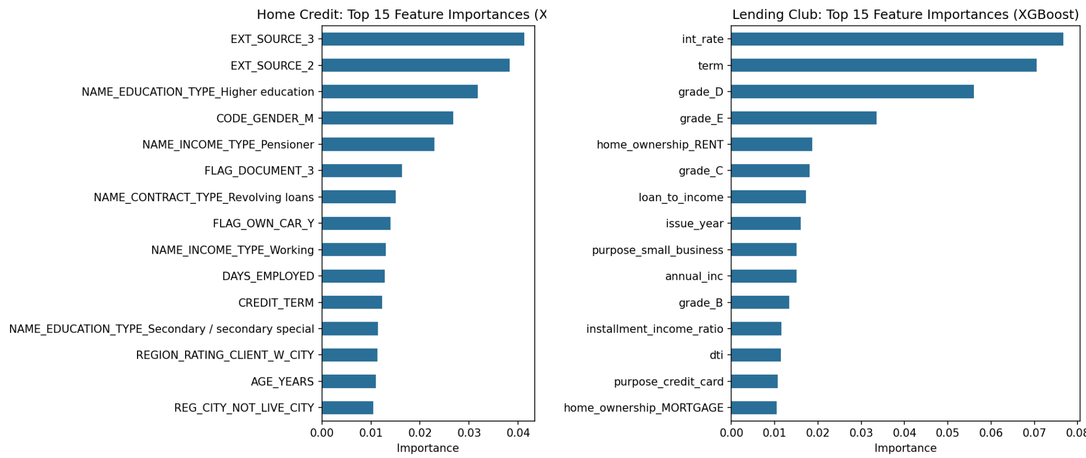

Fig. 7. Top 15 XGBoost feature importances for Home Credit (left) and Lending Club (right).

3.5 Insights and Complementarity

Both data sources produced usable, statistically sound insight: neither dataset failed to yield a discriminative model, with both achieving ROC-AUC comfortably above the 0.5 random-chance baseline. The insights are **complementary rather than contradictory**. Feature importance rankings differ substantially between the two lenders (Figure 7): Home Credit's top predictors are dominated by external bureau scores and applicant demographics (age, employment tenure), while Lending Club's top predictors are dominated by loan pricing and utilisation variables (interest rate, revolving utilisation, debt-to-income ratio). This divergence is expected and informative rather than conflicting: it reflects differences in what information each lender collects and prices at origination, not a disagreement about the underlying phenomenon of default risk. Both models converge on the same higher-level conclusion, that debt-servicing burden relative to income and externally verified creditworthiness are the dominant drivers of default, which is consistent with established credit risk theory and reinforces confidence in the modelling approach rather than undermining it.

3.6 Limitations and Future Improvements

Several limitations should be acknowledged. First, absolute precision remains low (0.17–0.31) at the default 0.5 classification threshold; in a production setting, this threshold would be tuned against the business cost of a missed default versus a false decline, using the precision-recall curve in Figure 6 rather than a fixed default cut-off. Second, this analysis used a single train/test split rather than cross-validation, which would give a more robust estimate of generalisation error and is a natural next step. Third, the Home Credit dataset was modelled using only the primary application table; the competition's full data also includes bureau, previous-application, and instalment-payment history tables which, if joined, would likely improve predictive performance further. Finally, hyperparameters were set to sensible defaults informed by common practice rather than exhaustively tuned via grid or Bayesian search, which represents a further avenue for improvement.

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Generative AI Attribution and Declaration

I used RMIT VAL to search for and review current data science job opportunities and to understand the current employment market relevant to the role selected for this assignment. I also used ChatGPT to help generate and refine ideas related to the selected job, credit-risk data science applications, and the structure and wording of some sections. The final selection of the job and datasets, analysis, interpretation of results, and submission decisions were reviewed and completed by me.

The AI attribution record was prepared using the AI Attribution Toolkit.

I certify that the submission of this assignment is incomplete without the submission of the completed **Condition 3 Bounded Process AI Declaration Form** (templates available in the specification of the assignment on Canvas).

Acknowledgments

References

[1] Home Credit Group. 2018. Home Credit Default Risk. Kaggle competition. https://www.kaggle.com/competitions/home-credit-default-risk/overview

A Job Listing

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Analysis & Reporting (Banking & Financial Services)

Full time

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Our client is a fast growing Australian fintech and due to growth, the team are expanding their existing Data Science function with a new hire to the team.

The Opportunity

As a Senior Data Scientist, you'll drive meaningful change by identifying key trends and building the automated decision systems that power credit and fraud risk management, with the ability to move from theory to execution quickly and see your work make a tangible, visible impact.

Reporting to the Head of Data and working within the risk team, you'll bring structure, insight and clarity to complex challenges, working cross-functionally with Growth, Operations and Product.

What you'll do

- Design and develop credit and fraud models across acquisition, lifecycle and collections optimisation that power the risk systems
- Build and deploy machine learning models end-to-end, leveraging cutting-edge AI/ML solutions to optimise and improve existing systems
- Maintain financial risk strategies in production, deploying updates and improvements driven by data analysis
- Collaborate closely with Growth, Operations and Product on financial risk decisions and processes

What success looks like

- Designing, developing and deploying machine learning and AI solutions with clear, measurable business impact
- Uplifting and improving the risk system through cutting-edge AI/ML applications
- Building and improving risk systems that contribute to the overall health of the wider ecosystem

What you'll bring

- Degree in Data Science, Computer Science, Statistics, Mathematics or a related field
- Strong Python skills with hands-on experience building and deploying machine learning models end-to-end in production
- Background in financial risk, collections optimisation or acquisition modelling is highly regarded, but candidates from other data-intensive backgrounds are genuinely welcome to apply
- Proficiency in Python, SQL and machine learning systems
- An understanding of, or strong interest in, risk management principles, payment processing systems and collections strategies
- Excellent communication skills, with the ability to collaborate effectively across functional teams and stakeholders
- Self-motivated and autonomous, with strong support from an established team

Please note this role is Melbourne based and requires candidates to be in the office three days per week.

To find out more about this confidential opportunity, apply now or reach out to Marie Thow at ••••@bluefinresources.com.au for a confidential conversation.

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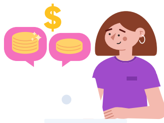

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SEEK acknowledges the Traditional Custodians of the lands on which it operates its online employment marketplace. The gum leaf represents a symbol of welcome and it acknowledges the diverse countries, environments and communities.

Artwork by Bitja, Dixon Patten Jnr, Gunnai, Gunditjmara, Yorta Yorta and Dhudhuroa, Bayila Creative.

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B Cover Letter

Dhrumil Ashish Parikh
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Hiring Manager

Dear Hiring Manager,

I would like to apply for the Data Scientist role listed by Bluefin Resources Pty Limited on behalf of its fintech client in consumer lending. Currently, I am pursuing the Master's program in Data Science at RMIT University, where I am developing skills in data analytics, machine learning, and data-driven decision-making, with a growing interest in credit risk.

Previously, I worked with NIQ in India, a global consumer intelligence company. This experience gave me practical exposure to data quality, data analysis, reporting, and transforming information into useful business insights. Working with real business data also helped me develop a practical approach to solving data-related problems.

In addition to my professional experience, I have developed skills in Python, SQL, data preprocessing, exploratory data analysis, and machine learning. My current data science project focuses on predicting loan repayment risk using two publicly available credit datasets. The aim is to identify patterns associated with repayment risk and evaluate machine learning models for credit-risk decision-making.

I am particularly interested in this opportunity because it combines technical modelling with collaboration across risk, engineering, and product teams. Learning, teamwork, effective communication, and responsible use of data are important to me, and I would value the opportunity to apply these principles in a fintech environment.

I am currently based in Melbourne and keen to continue my data science career in the financial technology sector.

Thank you for considering my application.

Yours sincerely,

Dhrumil Ashish Parikh

GitHub Repository

GitHub Repository